



EENG 385 - Electronic Devices and Circuits
PID Controller
Lab Handout

Outcome and Objectives

The PID Controller board is designed as a practical laboratory experience for students to understand how to characterize a plant, and tune a PID Controller. Through this process you will achieve the following learning objectives:

- Provide students will learn the “feel” of how PID control works by changing the P, I, D terms of the controller and noting the output.
- Building a Bode plot for a plant (low pass filter).
- Build a Nyquist plot for a plant (low pass filter).
- Using the Nyquist plot, predict the gain at which the system goes unstable.

The PID Controller Board

A high-level overview of the PID Controller board is shown in Figure 1. The input can come from a push-button (step input), a potentiometer (analog input), or a function generator using a user selectable jumper (high tech, I know). The K_p , K_i and K_d terms are adjustable from 0 to infinity using a potentiometer with a long stem. The control signal is converted into a PWM signal whose duty cycle is proportional to the control signal. This PWM signal is then converted into an analog output using a low pass filter.

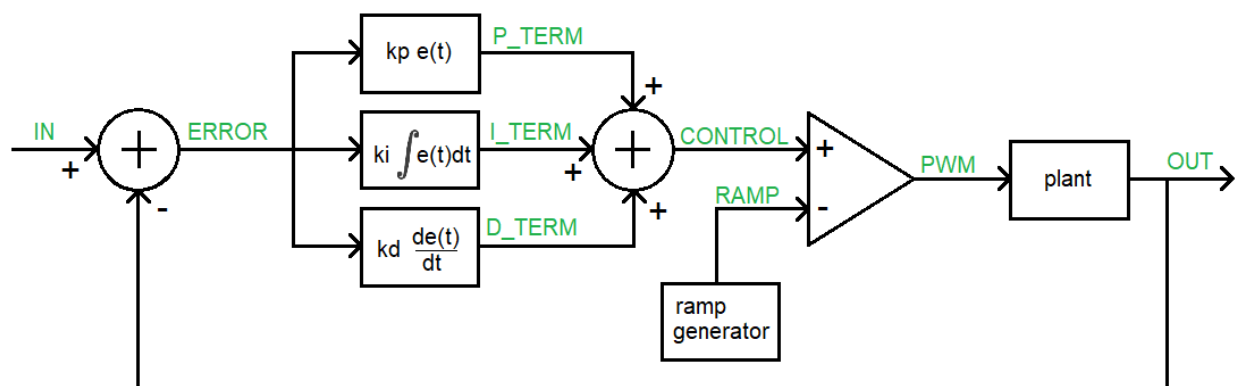


Figure 1: High-level architecture of the PID Controller board.

Importantly, the input and output of the plant can be isolated from the controller circuit. This will allow you to characterize the plant using the FRA capability of our oscilloscopes. Well now that you have an idea of what you are working with, let's talk more about the plant.

Circuit Operation:

Your task is to identify the building blocks in the schematic, analyze their behavior and relate these building blocks to the PID controller shown in **Error! Reference source not found.** When referencing opamps in the schematic, use the “ICxx” label, not the “TLC274P” label. When needed, assume that VCC=9V and VSS=-9V. Annoying to setup but this allows the error to be either positive or negative according to the sign of the voltage – very usefully semantically.

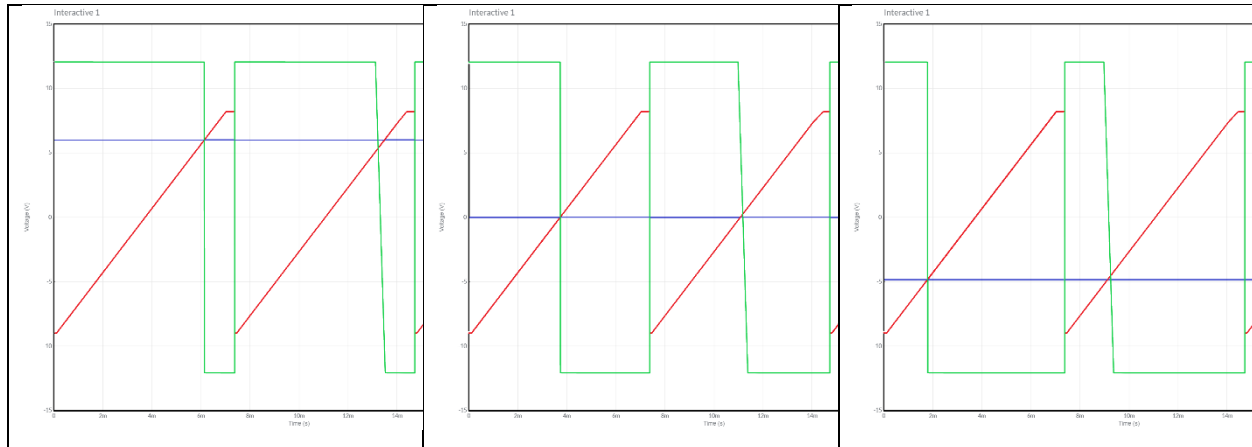
- 1) The first summer junction
 - a. What opamp and resistors form the summer junction that outputs **ERROR** in **Error! Reference source not found.**
IC2B, R15, R16, R17, R18
 - b. Write an equation for the output **ERROR** in terms of **IN** and **OUT**.
ERROR = IN - OUT
- 2) The proportional term, **P_TERM**, is formed by two opamps. **ERROR** is the input to the first stage and **P_TERM** is the output from the second stage.
 - a. What opamp/resistors in the form the first stage of the **P_TERM** in **Error! Reference source not found.**
IC2C, R5
 - b. What is the range of the gain of this circuit? Make sure to include the sign.
This is an inverting configuration so Gain = - R2/R2 [0, -∞]
 - c. What opamp/resistors form the second stage of the **P_TERM** in **Error! Reference source not found.**
IC2D, R8, R10
 - d. What is the gain of this second stage?
This is an inverting configuration so Gain = - R2/R2 = -1
 - e. What is the purpose of this second opamp stage?
Invert the gain from the first stage so the overall gain is positive.
 - f. Why didn't I use a non-inverting opamp instead of this 2-stage approach? Hint, look at your answer for part b.
Because the gain of a non-inverting configuration is 1+R2/R1, meaning that you can only form gains which are greater or equal to 1 and perhaps you would want a gain that is less than 1.
- 3) The integral term, **I_TERM**, is formed by two opamps. **ERROR** is the input to the first stage and **I_TERM** is the output from the second stage.
 - a. What opamp/resistors in the form the first stage of the **I_TERM** in **Error! Reference source not found.**
IC1D, R6
 - b. Where have you seen this circuit before?
This is the same circuit as P_TERM circuit.
 - c. What opamp/resistors form the second stage of the **I_TERM** in **Error! Reference source not found.**
IC2D, R9, C4
 - d. Write an equation for the output of the second stage, **I_TERM**, in terms of the input (call it **ERROR**) and C and R. Use C and R in your equation, not their values of uF and 10kΩ.
I_TERM = 1/RC * ∫ERROR(t)

- e. Why did I use an inverting integrator circuit instead of a non-inverting integrator?
So that you could have a gain in the range $[0, -\infty]$.
- 4) The derivative term, **D_TERM**, is formed by two opamps. **ERROR** is the input to the first stage and **D_TERM** is the output from the second stage.
 - a. What opamp/resistors in the form the first stage of the **D_TERM** in **Error!**
Reference source not found.?
IC1C, R7
 - b. Where have you seen this circuit before?
This is the same circuit as P_TERM circuit.
 - c. What opamp/resistors form the second stage of the **D_TERM** in **Error! Reference source not found.?**
IC1B, R12, C3
 - d. Write an equation for the output of the second stage, **D_TERM**, in terms of the input (call it **ERROR**) and C and R. Use C and R in your equation, not their values of 1uF and 10kΩ.
 $D_TERM = 1/RC * dERROR(t)/dt$
 - e. Why did I use an inverting differentiator circuit instead of a non-inverting differentiator?
So that you could have a gain in the range $[0, -\infty]$.
- 5) The second summer junction
 - a. What opamp and resistors form the summer junction that outputs **CONTROL** in **Error! Reference source not found.?**
IC3A, R19, R20, R21, R22, R24
 - b. Write an equation for the output **CONTROL** in terms of **P_TERM**, **I_TERM** and **D_TERM**.
 $CONTROL = (P_TERM + I_TERM + D_TERM) / 3.3$
 - c. What value “should” the 3.3kΩ have? Why did I choose 3.3kΩ instead of this value?
The value should have been 3kΩ because you should divide by 3 because you added 3 terms.
 - d. Why did I use an non-inverting summer instead of the simpler inverting summer?
To make the control signal have the same sign as the inputs.
- 6) The ramp generator is uses the Schmitt Trigger Relaxation Oscillator that you built in Lab 2. Using your lab results will simplify answering the questions in this section.
 - a. What components form the Schmitt Trigger Relaxation Oscillator (STRO)?
IC3C, (optional R31) R32, R33, R34, R35, R36, R37, D1, D2, C11
 - b. What is the period/frequency and duty cycle of the waveform generated by the STRO? Assume that potentiometer R35 is set to 0Ω.
Period = 7ms, Frequency = 1/7ms = 143Hz with a duty cycle of 1%
 - c. Does R35 effect the time high or time low of the STRO waveform?
The potentiometer R35 effects the time low of the STRO waveform.
 - d. Components T1, T2, R29, and R30 forms a 0.7mA current source leaving the “bottom” of T1 (it’s collector). Assume that C10 is not populated, how long does it take to charge C9 from -9V to +9V? How would you describe the shape of the voltage vs. time waveform?
Since $I = CdV/dt$, then $dV/dt = I/C$ which equals $0.7mA/100nF = 7000V/sec$ In order to charge the capacitor from -9V to +9V, the capacitor would have to swing through 18V. So $18V * 1sec/7000V = 2.6mS$ Since dV/dt is a constant, the waveform will be a straight line (ramp) from -9V to +9V.

- e. What is the role of Q1 in generating the **RAMP** waveform?
Q1 periodically clears the charge off the capacitor pair C9, C10
- f. What is the role of R35 in this circuit?
Allows us to adjust the period of the reset signal to clear the charge off C9, C10.

7) The comparator

- a. Given **RAMP** (in red) and **CONTROL** (in blue) shown in the following timing diagrams, draw the output **PWM** signal using the open-loop opamp configuration shown in **Error! Reference source not found..**



- b. What is the relationship between the **CONTROL** signal and the duty cycle of the **PWM** signal? To determine this, plot the duty cycle vs control voltage and then write an equation to describe the straight-line $y=mx+b$ style.

$$\text{Duty cycle} = 100/18 \text{ CONTROL} + 50$$

5

Power Input

Use the 3 wires loops in the lower left corner of the PCB to connect power.

- The bottom loop, labeled VSS- is connected to -9V
- The middle loop labeled VSS+ is connected to +9V
- The upper loop labeled GND.1 is connected to GND

Each ON/OFF switches turns on +9V or -9V separately. I did this because we have SPDT switches in stock and do not have DPDT switches. Always throw both switches at the same time, either both ON or both off. This is simple to do with a single finger.

Input Selection

In the upper left corner of the board is a 2x3 male header with a green jumper installed. The green jumper can connect one of three inputs to the PID controller input

- 1) An analog input provided by the blue potentiometer just to the left of the header.
- 2) A step input actuated by the push button on the left edge of the board labeled "STEP"
- 3) A function generator that connects to the loops labeled FNC and GND.2

As shown in Figure 2 the jumper is connecting the function generator loop to the input of the PID controller.

Plant Selection

At the top right of the board is a pair of yellow jumpers. When installed, the PWM signal is the plant input and the plant output is routed to the OUT header. Removal of the headers allows you access to the plant to characterize it (draw its Bode plot).

Plant Gain

The plant gain is controlled by the blue potentiometer on the right side of the PCB and labeled PLANT GAIN. This is a single turn trimmer POT so will be very sensitive to small changes in the position. In FIGURE 1, the potentiometer is positioned just a bit shy of 1. You should adjust the POT so that the cross used to adjust the position of the center tp is as close to 1 as you can get.

PID Gains

The gain for each term is controlled by the Pgain, Igain or Dgain POTs in the top center of the board. When set to their mid position the gain of each term is 1. Full CCW corresponds to a gain of 0 and full CW, a gain of infinity.

Oscilloscope probe points

These are the set of unpopulated header pins at the top of the PCB. They are shown in FIGURE 4 above a block diagram of a PID controller. Each of the terms is a voltage that, with the exception of the PWM and RAMP signals, that represents the signed magnitude of the signal. So for example if the error is -1V, then the corresponding error signal is -1. When the plant is included in the feedback loop and the PID values tuned properly, you should see the OUT voltage track the SET voltage closely.

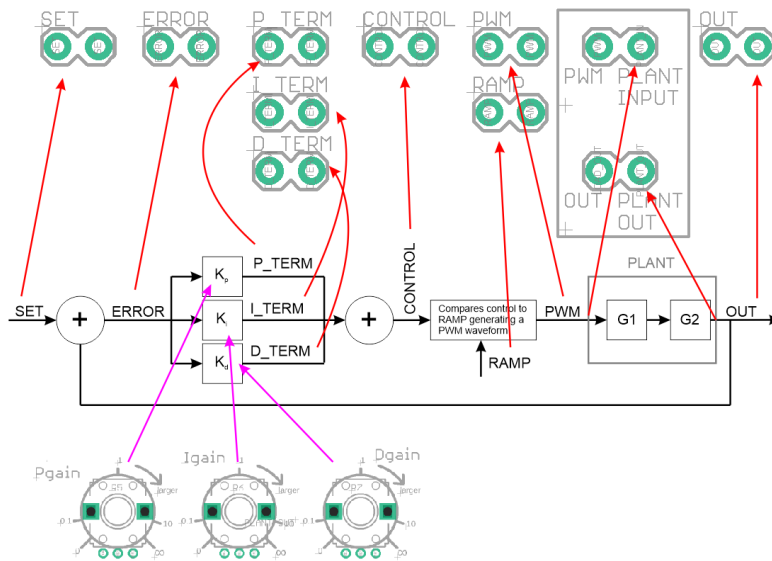
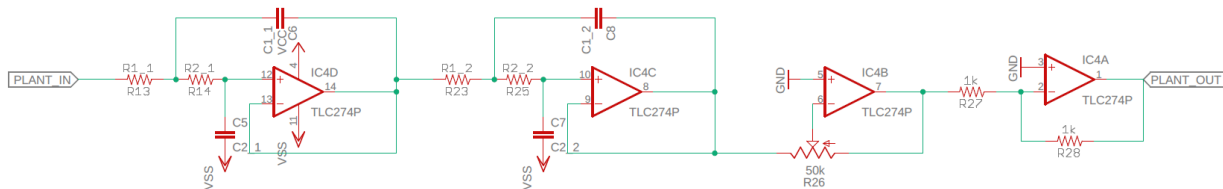


Figure 3: The headers at the top of the board provide access to the values of the PID control loop using an oscilloscope.

The Plant:

The plant of the PID Controller board is a 4th order low pass filter built using opamps IC4D/C. The gain of the plant is adjusted using the trimmer POT R26 attached to opamp IC4B. Since this opamp is inverting, the final opamp is an inverting configuration with gain -1.



In order to accommodate different plants the LPF components do not have fixed values, instead they use labels which are populated by components whose values create different filter types shown in the table below. Note, the filters were designed using the Analog Devices, Analog Filter Wizard using the following parameters.

Your board will have one of these four different amplifier configurations. Take a moment now, examine the components in your board and determine which LPF is populating your board and fill in their values in the table below. The capacitor values are given because they cannot be determined by inspection.

R1_1	R2_1	R1_2	R2_2	C1_1	C2_1	C1_2	C2_2
				100nF	68nF	100nF	10nF

Converting PWM to DC

The role of the plant is to convert the PWM output from the controller into a DC signal that is compared against the reference input. Mathematically you can see this by representing the PWM waveform by its Fourier Series. Consider this series to have two types of components, DC and AC.

The DC component of the Fourier Series is the average value of the PWM waveform. Since the frequency of DC is 0Hz the low pass filter passes it through with 0dB of attenuation. Since 0dB of attenuation is “no change”, the DC value passes unchanged through the filter.

The AC components of the PWM waveform have frequencies which are integer multiples of the fundamental frequency, which is the frequency of the PWM waveform, roughly 1kHz. These frequencies are a decade higher than the corner frequency, 100Hz, which means that their attenuation is at least -80dB. This is a substantial attenuation which will block these harmonic frequencies from appearing at the output.

Hence, the only input to reach the output is the DC which is the average value of the PWM waveform, exactly what we want.

Characterize The Plant:

The first thing that you will want to do is characterize the plant by making a Bode Plot. The goal of this will be to first adjust the plant gain so that it is 0dB in the pass-band and the second, to collect numerical data at this gain to build the Nyquist plot of the plant in Excel.

To do this you will:

- 1) Isolate the plant by removing the input and output jumpers.
- 2) Power on the oscilloscope, they take forever to boot.
- 3) Turn on and configure the power supply to have 9V, 0.1A on channel 1 and 2. Leave the power supply on, but **disable** the individual channels until everything is connected.
- 4) Configure the power supply to generate +9V, GND, -9V, attach the respective terminals to the PID Controller using red (+9V), green (GND) and black (-9V) banana cables and alligator clips.
- 5) Run a T-connected BNC from the oscilloscope function generator output to Channel 1 of the oscilloscope and to the PLANT_INPUT inputs on the PID Controller board.
- 6) Attach a probe to Channel 2 of the oscilloscope and connect it to the PLANT_OUT. No need for a ground clip, the function generator ground has already taken care of that.
- 7) Enable channel 1 and 2 of the power supply and ensure that you are not hitting the current limit on either channel. If so, **immediately disable both channels**. You have something incorrectly connected. Fix and try again.

Here is what your setup should look like. Yea, it's messy 😞

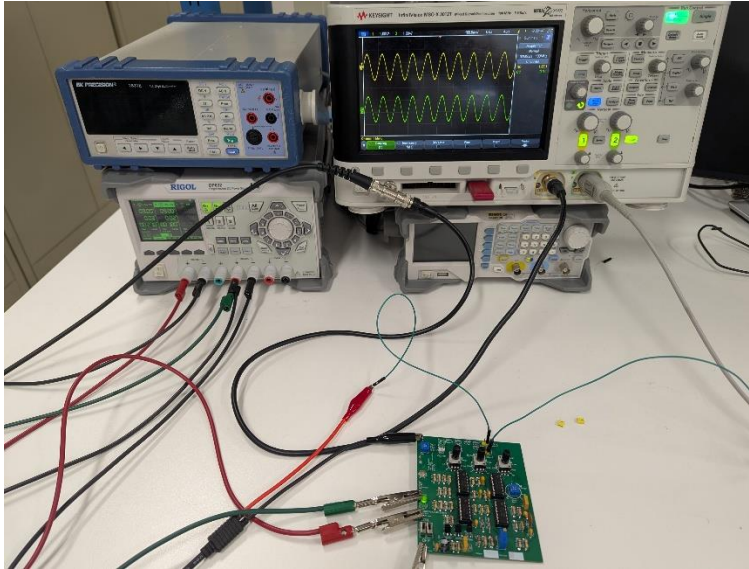


Figure 4: The PID Controller board configured to characterize the plant. Channel 2 is connected to the PLANT OUT off screen to the right.

Before running FRA on the oscilloscope, you should enable the function generator output for a frequency well inside the corner frequency, say 10Hz. Make sure that the input and output have the same amplitude (one probe is not 10x off), you should see a slight phase lag on the output like that in Figure 4. Now adjust the plant gain until the output has the same amplitude as the input, 0dB. Cool.

Before going on, examine the step response of the plant to a 4Hz 2V square wave. Without knowing that your plant is 4th order, would you characterize the plant's step response behavior as first order or second order? Why? Grab a screen shot of the step response to include in your lab report.

Now run the frequency response analysis on the oscilloscope from 10Hz to 1kHz. Your FRA must be close to 0dB through the pass band, 10Hz to 100Hz. If it is -20dB or +20dB then you have an attenuation issue with your scope probes. Figure 5 shows the plot that I got using a -1dB Chebyshev LPF.



Figure 5: The Bode plot from a -3dB Chebyshev LPF.

Now save the data used to generate the Bode plot on the oscilloscope by plugging in a thumb drive, pressing the SAVE/RECALL hard button, select the Format option, scroll to the bottom of the options, select **Frequency Response Analysis data (*.csv)** and then press the Press to Save softkey. You will use this data in the next section.

Creating a Bode plot

Note that Figure 6 shows additional columns H-K used to build the Nyquist plot in the next section. For now, ignore these columns.

- 1) Open the .csv file in Excel or import the .csv data into Excel.
- 2) Immediately save the file as a .xlsx file
- 3) Delete column C, it just tells you the amplitude of the voltage input.
- 4) Insert a new column E which converts any positive angles into the using the excel equation =IF(D2<0,D2,D2-360). You can type this equation into cell E2, hit enter, select cell E2, then click on the dot in the lower right corner of the cell and, while holding down the mouse button, drag the cell down, selecting all the rows with data in them. The equation should be copied into all these cells.

	A	B	C	D	E	F	G	H	I	J
1	#	Frequency	Gain (dB)	Phase (°)		Gain (V/V)		X	Y	Flip
2	1	10	2.62	-55.34		1.35207		0.76893	-1.11214	1.11214
3	2	10.8	2.03	-54.69		1.30317		0.75323	-1.06343	1.06343
4	3	11.7	2.03	-54.18		1.26328		0.73932	-1.02434	1.02434
5	4	12.6	1.76	-53.79		1.22462		0.72344	-0.98809	0.98809
6	5	13.6	1.54	-53.61		1.19399		0.70837	-0.96116	0.96116

Figure 6: The header of the Excel file used to generate the Bode and Nyquist plots.

- 5) Now make a Bode plot by selecting column B and C, then click Insert -> Scatter -> Scatter with Smooth lines. Make the horizontal axis logarithmic by carefully double clicking on one of the numerical values on the horizontal axis. In the Format Axis menu that appears on the right side of the screen, select logarithmic. I also like to change the minimum bound to 10 to fill the graph area. Nice. Do repeat the process with columns B and E to make the phase plot. Note the angle can jump around at higher frequencies. You are welcome to fix these few values by adding -360 degrees to them to smooth out the plot. Here is what I produced.

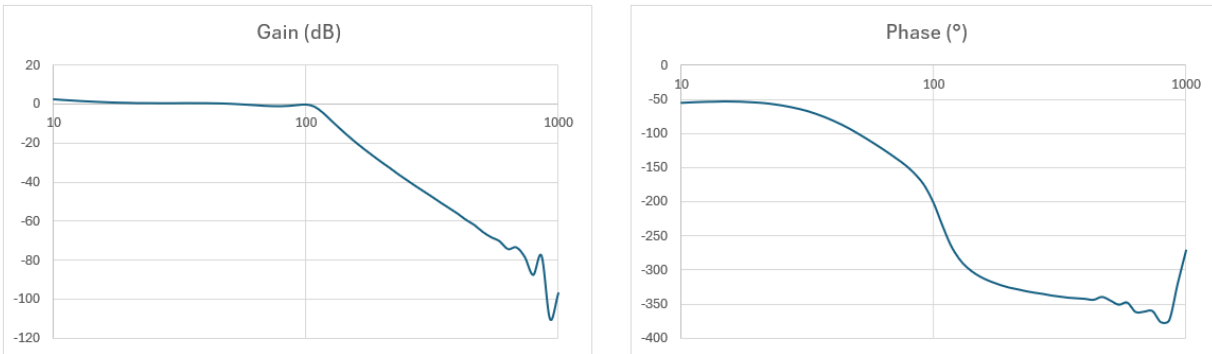


Figure 7: Bode plot for a -1dB Chebyshev LPF with a corner frequency of 100Hz.

Creating a Nyquist plot

- 1) Insert a header "Gain (V/V)" in column G. This column will contain the gain represented as a scalar value. To do this you will have to figure out the equation to convert the dB value in column C into a scalar.
- 2) Insert a header "X" in column I. This column will contain the x-coordinate of the complex vector that has a magnitude in the G column and a phase angle given in the E column. You will have to use a trig function to accomplish this.
- 3) Insert a header "Y" in column J. This column will contain the y-coordinate of the complex vector that has a magnitude in the G column and a phase angle given in the E column. You will have to use a trig function to accomplish this.
- 4) Select columns I and J and then Insert -> X Y (Scatter) -> Scatter with Smooth Lines. You now have half of the Nyquist plot. You can get the other half of the Nyquist plot by mirroring this half around the Y-axis. In other words multiply each Y coordinate by -1.
- 5) Insert a header "Flip" in column K. This column will contain the negative value in column J. After you populate this column, right click on you half Nyquist plot, click Select Data... In the Select Data Source pop-up, click Add. In the Edit Series pop-up, click in the Series X values box, then drag select all the X data in column I. Click in the Series Y values box, then drag select all the Flip data in column K. Click OK twice and your Nyquist plot should be complete.
- 6) Take a moment to adjust the X and Y scale of the plot by grabbing the lower right corner of the plot and stretch it so that the X and Y axis' have about the same scale size.

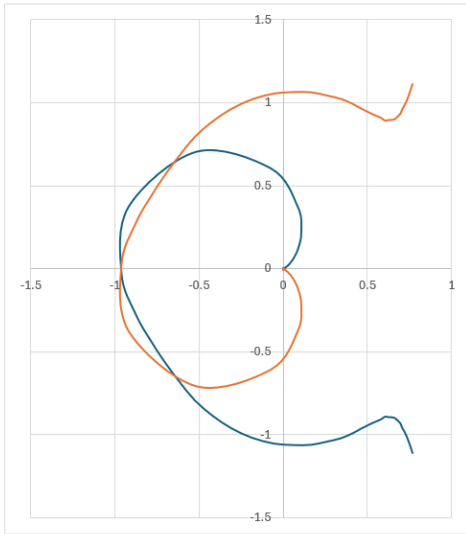


Figure 8: The Nyquist for a 1dB Chebyshev LPF.

Use the Nyquist plot to estimate the gain margin of your plant.

Tune the PID Controller:

Now, let's see if you can tune the PID controller so that the output of the plant can track a 4Hz square wave. To do this you will:

- 1) Insert the plant into the control loop by installing the jumpers on the PLANT INPUT and PLANT OUTPUT.
- 2) Power on the oscilloscope, they take forever to boot.
- 3) Turn on and configure the power supply to have 9V, 0.1A on channel 1 and 2. Leave the power supply on but **disable** the individual channels until everything is connected.
- 4) Configure the power supply to generate +9V, GND, -9V, attach the respective terminals to the PID Controller using red (+9V), green (GND) and black (-9V) banana cables and alligator clips.
- 5) Run a T-connected BNC from the oscilloscope function generator output to Channel 1 of the oscilloscope and to the FNC inputs on the PID Controller board. Attach the GND clip of the function generator to the nearby GND.2 loop.
- 6) Attach a probe to Channel 2 of the oscilloscope and connect it to the OUT header at the top of the PCB. No need for a ground clip, the function generator ground has already taken care of that.
- 7) Enable channel 1 and 2 of the power supply and ensure that you are not hitting the current limit on either channel. If so, **immediately disable both channels**. You have something incorrectly connected. Fix and try again.
- 8) Adjust the oscilloscope function generator to generate a 4Hz square wave with an amplitude of 1V.

- 9) Enable channel 2, make it the same scale as channel 1 and place the GND references of the two channels in the vertical center of the screen.

Now it's time to play with the PID knobs to get the output to track the input. Figure 9 shows my setup and tuned controller, you should try to do better.

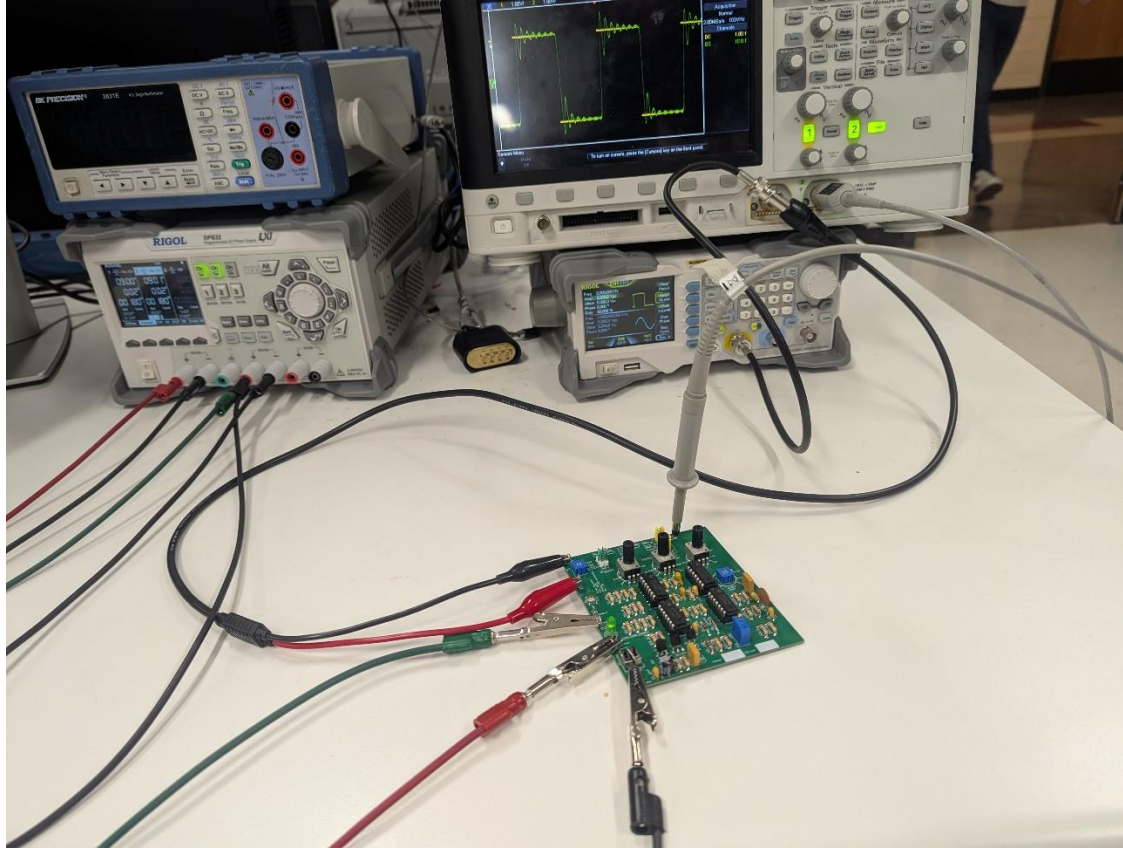


Figure 9: A tuned PID controller that allows the output of a 4th order Chebyshev LPF to track a square wave. Note, I used the function generator, you should probably just use the oscilloscope's function generator.

Screen shot the input/output of your tuned controller. Describe the function of the following controls in two sentences each.

- 1) The proportional gain
- 2) The integral gain
- 3) The derivative gain

Nyquist stability predicts the amount of plant gain that will cause a $K_p=1$ $K_i=0$ $K_d=0$ feedback controller to go unstable. Set your integral and derivative gain to 0 and your proportional gain to 1. My gain margin in Figure 8 was very small, meaning that only the slightest increase in the PLANT GAIN potentiometer should drive my output unstable.

Turn In:

- 1) Make a record of your response to numbered items below and turn in a single copy as your team's solution on Canvas using the instructions posted there.
- 2) Include the names of both team members at the top of your solutions.
- 3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

Circuit Operation

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Characterize The Plant

- Which plant you implemented on your board.
- An image of the Bode plot please include both magnitude and phase.
- The Nyquist plot.
- Step response of the plant with gain equal to 1.

Tune the PID Controller

For the tuned controller :

- An oscilloscope screenshot with the input on Ch1 and the output on Ch2 superimposed like that shown in Figure 9.
- The approximate values of K_p , K_i and K_d .
- An oscilloscope screenshot with Ch1 input and Ch2 error superimposed.
- An oscilloscope screenshot with Ch1 input and Ch2 proportional superimposed.
- An oscilloscope screenshot with Ch1 input and Ch2 integral superimposed.
- An oscilloscope screenshot with Ch1 input and Ch2 derivative superimposed.